

KHUSHI MAHESH

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EDUCATION

PES University — Bangalore, India

2023 – 2027

Bachelor of Technology in Computer Science, Specialization in Artificial Intelligence & Machine Learning — CGPA: 8.05, SGPA: 8.71

Distinction Scholarship, Computer Science Department (Semesters 3, 4, 5, 6)

EXPERIENCE

AI Intern — Yucca IT Solutions, Scalian Group

Jun 2026 – Present

- Engineered the core multi-agent orchestration pipeline using LangGraph and Neo4j for an AI project management platform, modeling complex SDLC workflows across 30+ specialized agents spanning project population, planning, and monitoring.
- Optimized LLM context efficiency and lowered API costs by designing targeted Cypher subgraph retrievals; replaced monolithic graph dumps with selective dependency scoping, reducing prompt token consumption by 40%.
- Implemented Human-in-the-Loop (HITL) state persistence leveraging LangGraph interrupts and SQLite versioning, enabling interactive user feedback loops, durable checkpoint rollbacks, and safe database commits for PM approvals.

Summer Research Intern — Center for Cloud Computing and Big Data (CCBD), PES University

Jun 2025 – Aug 2025

- Built a multi-agent LangGraph pipeline that transformed unstructured resumes into structured career profiles, integrating a Python data-ingestion framework with semantic retrieval across 4 external career datasets to support trajectory-aware career analysis.
- Engineered a hybrid recommendation engine using Neo4j, vector search, and retrieval-augmented generation (RAG) to identify skill gaps, retrieve evidence-backed career trajectories, and generate personalized career recommendations through graph-guided retrieval.
- Designed and implemented Graph-Grounded Multi-Agent Debate (GG-MAD), enabling specialized LLM agents to critique, refine, and validate career recommendations, with evaluation across 50 career profiles demonstrating improved recommendation quality over single-agent and vanilla debate baselines.

PROJECTS

Nrittam — AI-Driven Classical Dance Feedback Engine

2024 – 2025

- Built a computer-vision feedback system (OpenCV, MediaPipe) that analyzes Bharatanatyam form in real time and gives corrective feedback for self-assessment.
- Designed a gamified freemium model aligned with UNSDGs 4 and 10; placed 1st among 40 teams at the State-Level Pitchfest (National Startup Day) and was a Top 10 National Finalist at Cisco thingQbator Cohort 7 (NASSCOM Foundation).

Distributed File System & High-Availability Storage Engine

Aug 2025 – Nov 2025

- Built an HDFS-inspired distributed storage engine in Python with a multi-threaded Namenode for metadata coordination and heartbeat-driven Datanode health checks.
- Implemented 2MB file-chunking and replication achieving 100% data recoverability with zero single points of failure; built a real-time dashboard visualizing shard distribution and node health.

AirEase — AI-Integrated Wearable for Hyperhidrosis

Feb 2024 – Present

- Built a full-stack medical wearable prototype (ESP32, C/C++, React Native, Firebase) with low-power sensor-fusion firmware and a mobile app for real-time physiological monitoring and airflow-based cooling control.
- Top 25 National Finalist, Cisco thingQbator Cohort 7; Special Jury Mention, CellVerse Hackathon (winner among 24 teams).

FinFlow — Multilingual AI Financial Advisor

Mar 2025

- Built a guided loan-advisory tool for regional languages using Sarvam AI for language understanding and FAISS-based vector search for retrieval.
- Top 30 Finalist nationally (Great Bengaluru Hackathon 2025) in a highly competitive track.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, Java

AI/ML & LLM Systems: LangGraph, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), Pydantic, Model Context Protocol, Computer Vision (OpenCV, MediaPipe), NLP, Deep Learning, PyTorch

Data & Infrastructure: Neo4j, Knowledge Graphs, Vector Databases, Distributed Systems, REST APIs, Flask, TCP Sockets, Multi-threading

Tools & Platforms: Git/GitHub, Firebase, React Native, Ollama, Docker

ACHIEVEMENTS & CERTIFICATIONS

- Winner, State-Level Pitchfest, National Startup Day & Top 10 National Finalist, Cisco thingQbator Cohort 7 (NASSCOM Foundation) — Nrittam
- 2nd Runner-Up, Kodikon 5.0 National Hackathon
- Top 30 Finalist, Great Bengaluru Hackathon 2025 — FinFlow
- Coursera: Neural Networks and Deep Learning; Improving Deep Neural Networks (Hyperparameter Tuning, Regularization, Optimization); Structuring Machine Learning Projects

LEADERSHIP

- Head, Pixels Photography Club, PES University (2025–2026) — Managed a 40+ member team; led visual documentation for 100+ campus and festival events.
- Inner Circle Member, E-Cell (2024–2025) — Coordinated startup-focused workshops and entrepreneurial initiatives.